Math 231A HW 8

Zih-Yu Hsieh

December 01, 2025

1 D

Problem 1

Etingof Problem Set 5.1:

(1) Let V be a representation of $\mathfrak{g}$ and $W \subset V$ be a subrepresentation. Then $B_V = B_W + B_{V/W}$, where $B_V(x, y) = \operatorname{tr}(\rho_V(x)\rho_V(y))$.

(2) Let $I \subset \mathfrak{g}$ be an ideal. Then the restriction of the Killing form of $\mathfrak{g}$ to I coincides with the Killing form of I.

Solution:

(1) First, fix a basis $w_1, ..., w_k \in W$ (say $\dim W = k$), and extend it to an ordered basis of V, say $w_1, ..., w_k, v_1, ..., v_n \in V$ (say $\dim V = k + n$). Then, for all $x \in \mathfrak{g}$, $\rho_V(x)$ under this basis can be written as follow:

$$\mathcal{M}(\rho_V(x)) = \begin{pmatrix} A_x & B_x \\ 0 & C_x \end{pmatrix} \tag{1.1}$$

Where A_x is a $k \times k$ matrix (with entries $(a_{ij})_{1 \leq i,j \leq k}$), C_x is an $n \times n$ matrix (with entries $(c_{ij'})_{1 \leq i,j' \leq n}$), and B_x is a $k \times n$ matrix (with entries $(b_{i'j'})_{\substack{1 \leq i' \leq k \\ 1 \leq j' \leq n}}$). The reason is simply because $\rho_V(x)(W) \subseteq W$ by the definition of subrepresentation.

Then, notice that $A_x = \rho_W(x)$ under the basis $w_1, ..., w_k \in W$ by definition (since each $j \in \{1, ..., k\}$ satisfies $\rho_W(x) = \rho_V(x)(w_j) = \sum_{i=1}^k a_{ij} w_j$).

Also, $C_x = \rho_{V/W}(x)$ under the basis $\overline{v}_1, ..., \overline{v}_n \in V/W$ (where $\overline{v}_j$ denotes the image of v_j under the quotient $V \twoheadrightarrow V/W$), since for all $j' \in \{1, ..., n\}$, the matrix representation satisfies the following:

$$\rho_V(x)(v_{j'}) = \sum_{i'=1}^k b_{i'j'} w_{i'} + \sum_{i=1}^n c_{ij'} v_i \tag{1.2}$$

Which, under the quotient $\sum_{i'=1}^k b_{i'j'} w_{i'} \sim 0$, so we get $\rho_{V/W}(\overline{v}_{j'}) = \overline{\sum_{i=1}^n c_{ij'} v_i} = \sum_{i=1}^n c_{ij'} v_i$, showing under basis $\overline{v_1}, ..., \overline{v_n} \in V/W$, matrix of $\rho_{V/W}(x)$ is given by $C_x = (c_{ij'})_{1 \leq i,j' \leq n}$.

So, as a result, one has the following:

$$\begin{aligned}\forall x,y\in\mathfrak{g}, \mathcal{M}(\rho_V(x)\rho_V(y)) &= \mathcal{M}(\rho(V)(x))\cdot\mathcal{M}(\rho_V(y))\\ &= \begin{pmatrix}A_x & B_x\\ 0 & C_x\end{pmatrix}\begin{pmatrix}A_y & B_y\\ 0 & C_y\end{pmatrix}\\ &= \begin{pmatrix}A_xA_y & *\\ 0 & C_xC_y\end{pmatrix}\end{aligned} \tag{1.3}$$

Hence, $B_V(x,y) = \mathrm{tr}(\rho_V(x)\rho_V(y)) = \mathrm{tr}\left(\begin{smallmatrix}A_xA_y & *\\ 0 & C_xC_y\end{smallmatrix}\right) = \mathrm{tr}\big(A_xA_y\big)+\mathrm{tr}\big(C_xC_y\big)$. However, since $A_x = \mathcal{M}(\rho_W(x))$ and $C_x = \mathcal{M}\big(\rho_{V/W}(x)\big)$ (under the chosen basis), we have $\mathrm{tr}(A_xA_y) = B_W(x,y)$, and $\mathrm{tr}(C_xC_y) = B_{V/W}(x,y)$. So, $B_V(x,y) = B_W(x,y)+B_{V/W}(x,y)$, or $B_V = B_W + B_{V/W}$.

(2) First, given any $a\in I$, any $x\in\mathfrak{g}$ satisfies ad $a(x) = [a,x]\in I$ (by definition of ideal). Hence, ad $a:\mathfrak{g}\to\mathfrak{g}$ has im(ad a) $\subseteq I$. Which, if composing with the projection $\pi:\mathfrak{g}\twoheadrightarrow\mathfrak{g}/I$, $\pi\circ$ ad $a:\mathfrak{g}\twoheadrightarrow\mathfrak{g}/I$ is a zero map, which factors through $\mathfrak{g}/I$ (say denotes as $\overline{\text{ad } a}:\mathfrak{g}/I\to\mathfrak{g}/I$, this is a zero map).

So, similar to the previous section, let $W := I$, $V := \mathfrak{g}$, and $\rho := \text{ad}:\mathfrak{g}\to\mathfrak{gl}(\mathfrak{g})$, for any $a,b\in I$, one has the killing form on $\mathfrak{g},\mathfrak{g}/I$ satisfies $K_{\mathfrak{g}}(a,b) = K_I(a,b)+K_{\mathfrak{g}/I}(a,b)$. However, we've shown that any $a\in I$ has its action on $\mathfrak{g}/I$ being a zero map, hence $K_{\mathfrak{g}/I}(a,b) = 0$ (since it's trace of products of zero maps). So, $K_{\mathfrak{g}}(a,b) = K_I(a,b)$, showing the Killing form of I coincides with the Killing form of $\mathfrak{g}$ restricting to I.

2 D

Problem 2

Etingof Problem Set 5.2:

Show that for $\mathfrak{g} = \mathfrak{sl}(n,\mathbb{C})$, the Killing form is given by $K(x,y) = 2n\,\mathrm{tr}(xy)$.

Solution: First, recall that $\mathfrak{sl}(n,\mathbb{C})$ is simple, hence by **Problem 6** of this problem set (which will be proven), any invariant bilinear form is up to a scalar factor.

Given any $x,y,z\in\mathfrak{sl}(n,\mathbb{C})$, notice that $\mathrm{tr}(x,y)$ satisfies the following:

$$\mathrm{tr}([x,z]y) = \mathrm{tr}(xzy - zxy) = \mathrm{tr}(xzy - xyz) = \mathrm{tr}(x[z,y]) \tag{2.1}$$

Hence, tr is an invariant bilinear form, so its relation with the Killing form is up to a scalar factor.

Now, consider the element $E_{11}-E_{22}\in\mathfrak{sl}(n,\mathbb{C})$: First, we have the following:

$$\mathrm{tr}\Big((E_{11}-E_{22})^2\Big) = \mathrm{tr}(E_{11}+E_{22}) = 2 \tag{2.2}$$

Now, if compute $K(E_{11}-E_{22},E_{11}-E_{22})$, we first need to compute the adjoint action of it on the basis elements: Since $\mathfrak{sl}(n,\mathbb{C})$ has basis being all E_{kl} (where $k\neq l$) and $E_{kk}-E_{k+1,k+1}$ for integer $1\leq k<n$, then we have the following relations:

$$\mathrm{ad}(E_{11}-E_{22})(E_{kl}) = \begin{cases} 2E_{12} & k=1, l=2 \\ -2E_{21} & k=2, l=1 \\ E_{1l} & k=1, l\neq 1,2 \\ -E_{k1} & l=1, k\neq 1,2, \\ E_{2l} & k=2, l\neq 1,2 \\ -E_{k2} & l=2, k\neq 1,2 \\ 0 & \text{otherwise} \end{cases} \quad \mathrm{ad}(E_{11}-E_{22})\big(E_{kk}-E_{k+1,k+1}\big)=0 \tag{2.3}$$

Which, we have the following for square:

$$\mathrm{ad}(E_{11}-E_{22})^2(E_{kl}) = \begin{cases} 4E_{12} & k=1, l=2 \\ 4E_{21} & k=2, l=1 \\ E_{1l} & k=1, l\neq 1,2 \\ E_{k1} & l=1, k\neq 1,2, \\ E_{2l} & k=2, l\neq 1,2 \\ E_{k2} & l=2, k\neq 1,2 \\ 0 & \text{otherwise} \end{cases} \quad \mathrm{ad}(E_{11}-E_{22})^2\big(E_{kk}-E_{k+1,k+1}\big)=0 \tag{2.4}$$

Hence, for $\mathrm{ad}(E_{11}-E_{22})^2$, there are two basis elements being eigenvectors with eigenvalue 4 (namely E_{12}, E_{21}), and there are $4(n-2)$ basis elements being eigenvectors with eigenvalue 1 (namely all E_{kl}, where $k=1$ or $k=2$ while $l\neq 1,2$, or $l=1$ or $l=2$ while $k\neq 1,2$), and the rest are eigenvectors with eigenvalue 0 (total of $(n^2-1)-4(n-2)-2 = n^2-4n+5$ being eigenvector of 0).

Therefore, under this basis, we have $\mathrm{tr}\Big(\mathrm{ad}(E_{11}-E_{22})^2\Big) = 2\cdot 4 + 4(n-2)\cdot 1 + (n^2-4n+5)\cdot 0 = 4n$. So, this shows the following:

$$K(E_{11}-E_{22}, E_{11}-E_{22}) = \mathrm{tr}\Big(\mathrm{ad}(E_{11}-E_{22})^2\Big) = 4n = 2n\cdot\mathrm{tr}\Big((E_{11}-E_{22})^2\Big) \tag{2.5}$$

Since $K(x,y)$ is a scalar of $\mathrm{tr}(xy)$ (or vice versa), this indicates that $K(x,y) = 2n\cdot\mathrm{tr}(xy)$.

3 D

Problem 3

Etingof Problem Set 5.3:
Let $\mathfrak{g} \subset \mathfrak{gl}(n, \mathbb{C})$ be the subspace consisting of block-triangular matrices:

$$\mathfrak{g} = \left\{ \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} \right\} \tag{3.1}$$

Where A is a $k \times k$ matrix, B is a $k \times (n-k)$ matrix, and D is a $(n-k) \times (n-k)$ matrix.

(1) Show that $\mathfrak{g}$ is a Lie subalgebra (this is a special case of so-called *parabolic subalgebras*).

(2) Show that radical of $\mathfrak{g}$ consists of matrices of the form $\begin{pmatrix} \lambda \cdot I & B \\ 0 & \mu \cdot I \end{pmatrix}$, and describe $\mathfrak{g}/\mathrm{rad}(\mathfrak{g})$.

Solution:

(1) Given any $\begin{pmatrix} A & B \\ 0 & D \end{pmatrix}, \begin{pmatrix} A' & B' \\ 0 & D' \end{pmatrix} \in \mathfrak{g}$, one has the following:

$$\begin{aligned} \left[\begin{pmatrix} A & B \\ 0 & D \end{pmatrix}, \begin{pmatrix} A' & B' \\ 0 & D' \end{pmatrix} \right] &= \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} \begin{pmatrix} A' & B' \\ 0 & D' \end{pmatrix} - \begin{pmatrix} A' & B' \\ 0 & D' \end{pmatrix} \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} \\ &= \begin{pmatrix} AA' & * \\ 0 & DD' \end{pmatrix} - \begin{pmatrix} A'A & *' \\ 0 & D'D \end{pmatrix} \\ &= \begin{pmatrix} [A, A'] & * - *' \\ 0 & [D, D'] \end{pmatrix} \in \mathfrak{g} \end{aligned} \tag{3.2}$$

Where, the brakets in the columns are the commutator of $\mathfrak{gl}(k, \mathbb{C})$ for the top left, and the commutator of $\mathfrak{gl}(n-k, \mathbb{C})$ for the bottom right.

Hence, $\mathfrak{g}$ is closed under commutator, showing it's a Lie subalgebra.

(2) First, let $J := \left\{ \begin{pmatrix} \lambda \cdot I & B' \\ 0 & \mu \cdot I \end{pmatrix} \right\}$ be the given subspace of $\mathfrak{g}$, given any $\begin{pmatrix} A & B \\ 0 & D \end{pmatrix} \in \mathfrak{g}$, one has the following:

$$\begin{aligned} \left[\begin{pmatrix} \lambda \cdot I & B' \\ 0 & \mu \cdot I \end{pmatrix}, \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} \right] &= \begin{pmatrix} \lambda \cdot I & B' \\ 0 & \mu \cdot I \end{pmatrix} \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} - \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} \begin{pmatrix} \lambda \cdot I & B' \\ 0 & \mu \cdot I \end{pmatrix} \\ &= \begin{pmatrix} \lambda A & * \\ 0 & \mu D \end{pmatrix} - \begin{pmatrix} \lambda A & *' \\ 0 & \mu D \end{pmatrix} = \begin{pmatrix} 0 & * - *' \\ 0 & 0 \end{pmatrix} \in I \end{aligned} \tag{3.3}$$

Hence, this shows that J is an ideal of $\mathfrak{g}$.

Also, notice that $J \subset \mathfrak{ut}(n, \mathbb{C})$ (where $\mathfrak{ut}(n, \mathbb{C})$ denotes the Lie subalgebra of upper-triangular matrices), since $\mathfrak{ut}(n, \mathbb{C})$ is solvable, then J is solvable. So, one hsa $J \subseteq \mathrm{rad}(\mathfrak{g})$ by definition.

Now, to show that $J = \mathrm{rad}(\mathfrak{g})$, consider the two projections $p_1 : \mathfrak{g} \to \mathfrak{gl}(k, \mathbb{C})$ and $p_2 : \mathfrak{g} \to \mathfrak{gl}(n-k, \mathbb{C})$, by $p_1 \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} = A$, and $p_2 \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} = D$, which the two are clearly linear maps. However, based on the commutation relation in part (1), these two are also Lie-algebra homomorphisms:

$$\begin{aligned} p_1 \left[\begin{pmatrix} A & B \\ 0 & D \end{pmatrix}, \begin{pmatrix} A' & B' \\ 0 & D' \end{pmatrix} \right] &= p_1 \begin{pmatrix} [A, A'] & * - *' \\ 0 & [D, D'] \end{pmatrix} \\ &= [A, A'] = \left[p_1 \begin{pmatrix} A & B \\ 0 & D \end{pmatrix}, p_1 \begin{pmatrix} A' & B' \\ 0 & D' \end{pmatrix} \right] \end{aligned} \tag{3.4}$$

$$p_2\left[\begin{pmatrix} A & B \\ 0 & D \end{pmatrix}, \begin{pmatrix} A' & B' \\ 0 & D' \end{pmatrix}\right] = p_2\begin{pmatrix} [A, A'] & * - *' \\ 0 & [D, D'] \end{pmatrix}$$
$$= [D, D'] = \left[p_2\begin{pmatrix} A & B \\ 0 & D \end{pmatrix}, p_2\begin{pmatrix} A' & B' \\ 0 & D' \end{pmatrix}\right] \tag{3.5}$$

Hence, as a result, $p_1(\mathrm{rad}(\mathfrak{g})) \subseteq \mathfrak{gl}(k, \mathbb{C}), p_2(\mathrm{rad}(\mathfrak{g})) \subseteq \mathfrak{gl}(n-k, \mathbb{C})$ are both solvable ideals (since p_1, p_2 are surjective in this case), showing $p_1(\mathrm{rad}(\mathfrak{g})) \subseteq \mathrm{rad}(\mathfrak{gl}(k, \mathbb{C})) = \{\lambda \cdot I_k\}$, and $p_2(\mathrm{rad}(\mathfrak{g})) \subseteq \mathrm{rad}(\mathfrak{gl}(n-k, \mathbb{C})) = \{\mu \cdot I_{n-k}\}$ (since classical lie algebras are reductive, meaning $\mathrm{rad}(\mathfrak{gl}(l, \mathbb{C})) = \mathfrak{z}(\mathfrak{gl}(l, \mathbb{C})) = \{\lambda \cdot I_l\}$, or the radical and the center coincides).

Hence, $\mathrm{rad}(\mathfrak{g}) \subseteq p_1^{-1}\{\lambda \cdot I_k\} = \left\{\left(\begin{smallmatrix} \lambda\cdot I & B \\ 0 & D \end{smallmatrix}\right)\right\}$, and $\mathrm{rad}(\mathfrak{g}) \subseteq p_2^{-1}\{\mu \cdot I_{n-k}\} = \left\{\left(\begin{smallmatrix} A & B' \\ 0 & \mu\cdot I \end{smallmatrix}\right)\right\}$. Hence, $\mathrm{rad}(\mathfrak{g}) \subseteq \left\{\left(\begin{smallmatrix} \lambda\cdot I & B \\ 0 & D \end{smallmatrix}\right)\right\} \cap \left\{\left(\begin{smallmatrix} A & B' \\ 0 & \mu\cdot I \end{smallmatrix}\right)\right\} = \left\{\left(\begin{smallmatrix} \lambda\cdot I & B \\ 0 & \mu\cdot I \end{smallmatrix}\right)\right\} = J$. As a conclusion, $\mathrm{rad}(\mathfrak{g}) = J$.

Finally, to talk about the structure of $\mathfrak{g}/\mathrm{rad}(\mathfrak{g})$, it can be described by $\left\{\left(\begin{smallmatrix} A & 0 \\ 0 & D \end{smallmatrix}\right) \;\middle|\; A \in \mathfrak{sl}(k, \mathbb{C}), D \in \mathfrak{sl}(n-k, \mathbb{C})\right\} =: \mathfrak{k}$ (the block diagonals are contained in $\mathfrak{sl}$).

(Note: As a Lie-algebra, since all its components are in $\mathfrak{sl}$, and are block diagonal, we have $\mathfrak{k} \cong \mathfrak{sl}(k, \mathbb{C}) \oplus \mathfrak{sl}(n-k, \mathbb{C})$).

Given any $\left(\begin{smallmatrix} A & B \\ 0 & D \end{smallmatrix}\right) \in \mathfrak{g}$, take $\lambda := \frac{\mathrm{tr}(A)}{k}$, and $\mu := \frac{\mathrm{tr}(D)}{n-k}$, define $A' := A - \lambda \cdot I_k$ and $D' := D - \mu \cdot I_{n-k}$. Notice that $\mathrm{tr}(A') = \mathrm{tr}(A) - k \cdot \lambda = \mathrm{tr}(A) - \mathrm{tr}(A) = 0$, and $\mathrm{tr}(D') = \mathrm{tr}(D) - (n-k) \cdot \mu = \mathrm{tr}(D) - \mathrm{tr}(D) = 0$, showing $A' \in \mathfrak{sl}(k, \mathbb{C})$ and $D' \in \mathfrak{sl}(n-k, \mathbb{C})$. Where, the matrix $\left(\begin{smallmatrix} A' & 0 \\ 0 & D' \end{smallmatrix}\right) \in \mathfrak{k}$, one has the following:

$$\begin{pmatrix} A & B \\ 0 & D \end{pmatrix} - \begin{pmatrix} A' & 0 \\ 0 & D \end{pmatrix} = \begin{pmatrix} \lambda \cdot I_k & B \\ 0 & \mu \cdot I_{n-k} \end{pmatrix} \in \mathrm{rad}(\mathfrak{g}) \tag{3.6}$$

Hence, $\left(\begin{smallmatrix} A' & 0 \\ 0 & D' \end{smallmatrix}\right) \in \mathfrak{k}$ is a representative of $\left(\begin{smallmatrix} A & B \\ 0 & D \end{smallmatrix}\right)$ in $\mathfrak{g}/\mathrm{rad}(\mathfrak{g})$.

Now, suppose $\left(\begin{smallmatrix} A' & 0 \\ 0 & D' \end{smallmatrix}\right) \in \mathfrak{k}$ represents the zero coset in $\mathfrak{g}/\mathrm{rad}(\mathfrak{g})$, one has $\left(\begin{smallmatrix} A' & 0 \\ 0 & D' \end{smallmatrix}\right) \in \mathrm{rad}(\mathfrak{g})$, hence there exists $\lambda, \mu \in \mathbb{C}$, such that $A' = \lambda \cdot I_k$, and $D' = \lambda \cdot I_{n-k}$. However, since $A' \in \mathfrak{sl}(k, \mathbb{C})$ and $D' \in \mathfrak{sl}(n-k, \mathbb{C})$, one has $0 = \mathrm{tr}(A') = \mathrm{tr}(\lambda \cdot I_k) = k \cdot \lambda$ and $0 = \mathrm{tr}(D') = \mathrm{tr}(\mu \cdot I_{n-k}) = (n-k) \cdot \mu$. With $k, n-k \neq 0$ (for nontrivial case), then $\lambda, \mu = 0$. Hence, $A' = 0$ and $D' = 0$, showing $\left(\begin{smallmatrix} A' & 0 \\ 0 & D' \end{smallmatrix}\right) = 0$.

Hence, there is a linear one-to-one correspondance between $\mathfrak{sl}(k, \mathbb{C}) \oplus \mathfrak{sl}(n-k, \mathbb{C}) \cong \mathfrak{k}$ and $\mathfrak{g}/\mathrm{rad}(\mathfrak{g})$ (that also agrees on the Lie-algebra structure), hence $\mathfrak{g}/\mathrm{rad}(\mathfrak{g}) \cong \mathfrak{sl}(k, \mathbb{C}) \oplus \mathfrak{sl}(n-k, \mathbb{C})$.

4 D

Problem 4

Etingof Problem Set 5.4:
Show that the bilinear form $\mathrm{tr}(xy)$ on $\mathfrak{sp}(n, \mathbb{K})$ is non-degenerate.

Solution: (Here, assume $\mathrm{char}(\mathbb{K}) \neq 2$). First, recall that $\mathfrak{sp}(n, \mathbb{K})$ is the Lie algebra formed by all matrices $\left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$ (where $A, B, C, D \in \mathfrak{gl}(n, \mathbb{K})$), such that $D = -A^T$, $B = B^T$, and $C = C^T$. Which, it means the matrix A can be generated by the standard basis matrices $\left(E_{ij}\right)_{1 \leq i,j \leq n} \subset \mathfrak{gl}(n, \mathbb{K})$, while B, C must be generated by the pairs $\left(E_{ij} + E_{ji}\right)_{i \leq j}$ by the symmetric properties of B, C.

Hence, the generators of $\mathfrak{sp}(n, \mathbb{K})$ are of the following forms:

$$\begin{cases} \left(\begin{smallmatrix} E_{ij} & 0 \\ 0 & -E_{ji} \end{smallmatrix}\right) & 1 \leq i, j \leq n \\ \left(\begin{smallmatrix} 0 & 0 \\ E_{ij}+E_{ji} & 0 \end{smallmatrix}\right) & i \leq j \\ \left(\begin{smallmatrix} 0 & E_{ij}+E_{ji} \\ 0 & 0 \end{smallmatrix}\right) & i \leq j \end{cases} \tag{4.1}$$

Which, to check that $\mathrm{tr}(xy)$ is non-degenerate on $\mathfrak{sp}(n, \mathbb{K})$, it suffices to check it for the generators. (And, as a side note, recalll that $E_{ij}E_{kl} = \delta_{jk}E_{il}$, with δ_{jk} denotes the Kronecker Delta).

- For the first one, we have the following:

$$\begin{gathered} \begin{pmatrix} E_{ij} & 0 \\ 0 & -E_{ji} \end{pmatrix} \begin{pmatrix} E_{ji} & 0 \\ 0 & -E_{ij} \end{pmatrix} = \begin{pmatrix} E_{ij}E_{ji} & 0 \\ 0 & E_{ji}E_{ij} \end{pmatrix} = \begin{pmatrix} E_{ii} & 0 \\ 0 & E_{jj} \end{pmatrix} \\ \implies \mathrm{tr}\left(\begin{pmatrix} E_{ij} & 0 \\ 0 & -E_{ji} \end{pmatrix} \begin{pmatrix} E_{ji} & 0 \\ 0 & -E_{ij} \end{pmatrix} \right) = \mathrm{tr}(E_{ii}) + \mathrm{tr}\big(E_{jj}\big) = 2 \end{gathered} \tag{4.2}$$

$$\begin{gathered} \begin{pmatrix} 0 & 0 \\ E_{ij} + E_{ji} & 0 \end{pmatrix} \begin{pmatrix} 0 & E_{ij} + E_{ji} \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & (E_{ij} + E_{ji})^2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & \delta_{ji}E_{ij} + 1E_{ii} + \delta_{ij}E_{ji} + 1E_{jj} \end{pmatrix} \\ \implies \mathrm{tr}\left(\begin{pmatrix} 0 & 0 \\ E_{ij} + E_{ji} & 0 \end{pmatrix} \begin{pmatrix} 0 & E_{ij} + E_{ji} \\ 0 & 0 \end{pmatrix} \right) = 2 + \delta(ij) + \delta(ji) \neq 0 \end{gathered} \tag{4.3}$$

(Note: since $\delta_{ij} = \delta_{ji}$, the second value is either 2 or 4; under the assumption $\mathrm{char}(\mathbb{K}) \neq 2$ such value is never 0).

So, since all the generators of $\mathfrak{sp}(n, \mathbb{K})$ have a corresponding element also in $\mathfrak{sp}(n, \mathbb{K})$, such that the form $\mathrm{tr}(x, y) \neq 0$, then such form is non-degenerate.

5 D

Problem 5

Etingof Problem Set 5.5:

Let $\mathfrak{g}$ be a real Lie algebra with positive definite Killing form. Show that then $\mathfrak{g} = 0$. (Hint: $\mathfrak{g} \subset \mathfrak{so}(\mathfrak{g})$).

Solution: Suppose the contrary that as a reall Lie algebra $\mathfrak{g} \neq 0$, and it has a positive definite Killing form. Let $K : \mathfrak{g} \times \mathfrak{g} \to \mathbb{R}$ denotes the Killing form of $\mathfrak{g}$. Since the Killing form is positive definite (and is a real bilinear form), it satisfies all the properties of a real inner product, so one can choose $x_1, ..., x_n \in \mathfrak{g}$ to form an orthonormal basis with respect to the Killing form as inner product, then for any $x \in \mathfrak{g}$, one has $x = \sum_{i=1}^n K(x, x_i)x_i$.

Now, for all $x \in \mathfrak{g}$, consider $\text{ad } x \in \mathfrak{gl}(\mathfrak{g})$ with its matrix form with respect to the chosen orthonormal basis $x_1, ..., x_n$, denote as $\mathcal{M}(\text{ad } x) = (a_{ij})_{1 \leq i,j \leq n}$. Using the invariance of Killing form, for any x_i, x_j, one has the following:

$$-K(\text{ad } x(x_i), x_j) = K([x_i, x], x_j) = K(x_i, [x, x_j]) = K(x_i, \text{ad } x(x_j)) = K(\text{ad } x(x_j), x_i) \tag{5.1}$$

Where, since $\sum_{k=1}^n a_{ki}x_k = \text{ad } x(x_i) = \sum_{k=1}^n K(\text{ad } x(x_i), x_k)x_k$ (left side given by matrix entries, right side given by K as inner product), then $a_{ji} = K(\text{ad } x(x_i), x_j)$; similarly, one also has $a_{ij} = K(\text{ad } x(x_j), x_i)$. Then, based on the above equality, one has $a_{ij} = K(\text{ad } x(x_j), x_i) = -K(\text{ad } x(x_i), x_j) = -a_{ji}$, showing that matrix of $\text{ad } x$ is skew-symmetric, hence $\text{ad } x \in \mathfrak{so}(\mathfrak{g})$.

Then, notice that any operator $A \in \mathfrak{so}(\mathfrak{g})$ must have $\text{tr}(A^2) \leq 0$: If represent A under the chosen basis as above, since A is skew-symmetric, if let $c_1, ..., c_n$ denote the ordered column vectors of A, and $r_1, ..., r_n$ denote the ordered row vectors of A, it satisfies $r_i = -c_i^T$ (as matrices), hence one has the following:

$$\text{tr}(A^2) = \sum_{i=1}^n r_i c_i = \sum_{i=1}^n -c_i^T c_i = \sum_{i=1}^n -(c_i \cdot c_i) \leq 0 \tag{5.2}$$

Where, the last part is interpreting $c_i^T c_i$ (multiplication of row and column vector) as $c_i \cdot c_i$ (regular dot product on $\mathbb{R}^n$). Hence, for any $x \in \mathfrak{g}$, one also has $K(x, x) = \text{tr}((\text{ad } x)^2) \leq 0$ (since $\text{ad } x \in \mathfrak{so}(\mathfrak{g})$).

However, by positive definiteness, we must have $K(x, x) \geq 0$, together with the previous statement, this enforces $K(x, x) = 0$, which happens iff $x = 0$ (by positive definiteness). Hence, $\mathfrak{g} = 0$, which contradicts the assumption.

As a result, one must have $\mathfrak{g} = 0$.

6 D

Problem 6

Etingof Problem Set 5.6:

Let $\mathfrak{g}$ be a simple Lie algebra.

(1) Show that the invariant bilinear form is unique up to a factor.

(2) Show that $\mathfrak{g} \cong \mathfrak{g}^*$ as representations of $\mathfrak{g}$.

Solution: (Here, assume $\mathsf{char}(k) = 0$, and $k = \overline{k}$).1 In this case (2) is in fact easier to prove (while (1) follows from it), so we'll first show (2).

(2) To prove the statement that $\mathfrak{g} \cong \mathfrak{g}^*$ as representations of $\mathfrak{g}$ (where the first one is the adjoint representation, the second is the dual representation of that), it relies on two statements: First, the Killing form K is non-degenerate (since $\mathfrak{g}$ is simple, hence semisimple, showing $\ker(K) = 0$ by Cartan's semisimplicity criterion); also, K is invariant under adjoint action.

First, K is non-degenerate, meaning for all nonzero $x \in \mathfrak{g}$, one has $K(x, \_) : \mathfrak{g} \to k$ being a nonzero linear functional, hence, the linear map $f : \mathfrak{g} \to \mathfrak{g}^*$ by $f(x) = K(x, \_)$ is injective, while $\dim \mathfrak{g} = \dim \mathfrak{g}^*$, showing that such map f is indeed an isomorphism as vector spaces.

Now, to show it's a morphism of representations, consider the following: For any $x, y, z \in \mathfrak{g}$, one has $K(\text{ad } z(x), y) = K([z, x], y) = -K([x, z], y) = K(x, -[z, y]) = K(x, -\text{ad } z(y)) = K(x, \_) \circ (-\text{ ad } z)(y)$. Hence, one has $f(z \cdot x) = f(\text{ad } z(x)) = K(\text{ad } z(x), \_) = K(x, \_) \circ (-\text{ ad } z) = f(x) \circ (-\text{ ad } z) = z \cdot f(x)$, so f is a morphism of representation.

(Note: recall that if $\rho : \mathfrak{g} \to \mathfrak{gl}(V)$ is a Lie-algebra representation, then the dual representation $\rho^* : \mathfrak{g} \to \mathfrak{gl}(V^*)$ by $\rho^*(z) \cdot \varphi := \varphi \circ (-\rho(z))$).

So, $f : \mathfrak{g} \to \mathfrak{g}^*$ as both an isomorphism of vector spaces and a morphism of representation of $\mathfrak{g}$, then $\mathfrak{g} \cong \mathfrak{g}^*$ as representation of $\mathfrak{g}$.

(1) The goal for this part is to prove that $\mathfrak{g}$ as a representation of $\mathfrak{g}$, is irreducible (hence by Schur's Lemma $\text{Hom}_{\text{Lie}}(\mathfrak{g}, \mathfrak{g}^*) \cong k$, based on the assumption $k = \overline{k}$).

Suppose a subspace $\mathfrak{h} \subseteq \mathfrak{g}$ is a nontrivial subrepresentation of $\mathfrak{g}$, then for all $a \in \mathfrak{h}$ and $z \in \mathfrak{g}$, one has $z \cdot a = \text{ad } z(a) = [z, a] \in \mathfrak{h}$, showing that $[\mathfrak{g}, \mathfrak{h}] \subseteq \mathfrak{h}$, or $\mathfrak{h}$ is an ideal. But, based on the assumption of $\mathfrak{g}$'s simplicity, this enforces $\mathfrak{h} = \mathfrak{g}$. Hence, $\mathfrak{g}$ has no nontrivial proper subrepresentation, which is irreducible. Hence Schur's Lemma holds here.

Finally, notice that if $B : \mathfrak{g} \times \mathfrak{g} \to k$ is an invariant bilinear form, the linear map $g : \mathfrak{g} \to \mathfrak{g}^*$ by $g(x) = B(x, \_)$ is also a morphism of Lie-algebra representation, since every $x, y, z \in \mathfrak{g}$ again satisfy the following:

$$\begin{aligned} B(\text{ad } z(x), y) &= B([z, x], y) = -B([x, z], y) = -B(x, [z, y]) \\ &= B(x, -\text{ ad } z(y)) = B(x, \_) \circ (-\text{ad } z)(y) \end{aligned} \tag{6.1}$$

Hence, as linear maps, $g(z \cdot x) = g(\text{ad } z(x)) = B(\text{ad } z(x), \_) = B(x, \_) \circ (-\text{ ad } z) = g(x) \circ (-\text{ad } z) = z \cdot g(x)$, showing g is a morphsism of representation.

Hence, $g \in \text{Hom}_{\text{Lie}}(\mathfrak{g}, \mathfrak{g}^*) \cong k$, showing that $g = \lambda \cdot f$ for some $\lambda \in k$. Hence, for all $x, y \in \mathfrak{g}$, one has $B(x, y) = g(x)(y) = \lambda \cdot f(x)(y) = \lambda \cdot K(x, y)$, showing $B = \lambda \cdot K$, which is a scalar multiple of the Killing form of $\mathfrak{g}$.

So, all invariant bilinear form on $\mathfrak{g}$ is unique up to a factor in k.

7 D

Problem 7

Let V be a finite-dimensional complex vector space and let $A : V \to V$ be an upper-triangular operator. Let $F^k \subset \mathrm{End}(V)$, $-n \le k \le n$ be the subspace spanned by matrix units E_{ij} with $i - j \le k$. Show that then $\mathrm{ad}\, A \cdot F^k \subset F^{k-1}$ and thus, $\mathrm{ad}\, A : \mathrm{End}(V) \to \mathrm{End}(V)$ is nilpotent.

Solution:

(1) The statement is false in general:

For instance, choose $V = \mathbb{C}^2$, let $A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$, then $E_{21} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \in F^1$ in this case. However, if consider $\mathrm{ad}\, A(E_{21})$, we get:

$$\mathrm{ad}\, A(E_{21}) = AE_{21} - E_{21}A = \begin{pmatrix} 2 & 0 \\ 3 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 2 & -2 \end{pmatrix} \notin F^0 \tag{7.1}$$

Actually, a more precise requirement is A needs to be an upper-triangular operator with only one eigenvalue, i.e. under upper-triangular matrix form, the diagonal entries $A_{ii} = A_{jj}$ for all $i, j \in \{1, \dots, n\}$. We'll prove the statement for this assumption.

(2) The statement is true, if Diagonal Entries of A are the same:

Let $A = (a_{ij})_{1 \le i,j \le n}$ be an upper-triangular operator, with all the diagonal entries being the same (so, $a_{ii} = a_{jj}$ for all i, j, while $a_{ij} = 0$ for $i > j$). To prove $\mathrm{ad}\, A \cdot F^k \subset F^{k-1}$ for integer $-n \le k \le n$, it suffices to show that for all E_{ij} satisfying $i - j = k$, one has $\mathrm{ad}\, A(E_{ij}) \in F^{k-1}$ (since each $F^k = F^{k-1} \oplus \left(\bigoplus_{i-j=k} \mathbb{C} \cdot E_{ij}\right)$, so inductively if $\mathrm{ad}\, A \cdot F^{k-1} \subset F^{k-2} \subset F^{k-1}$, then for each E_{ij} with $i - j = k$, $\mathrm{Ad}\, A(E_{ij}) \in F^{k-1}$ guarantees $\mathrm{Ad}\, A \cdot F^k \subset F^{k-1}$, since each direct summand is mapped into F^{k-1}).

Now, fix an integer $-n \le k \le n$, for each E_{ij} satisfying $i - j = k$, one has the following:

$$\mathrm{ad}\, A(E_{ij}) = AE_{ij} - E_{ij}A = \begin{pmatrix} 0 & 0 & a_{1i} & 0 & 0 \\ 0 & 0 & \vdots & 0 & 0 \\ 0 & 0 & a_{ii} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & a_{jj} & \dots & a_{jn} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & a_{1i} & 0 & 0 \\ 0 & 0 & \vdots & 0 & 0 \\ 0 & 0 & 0 & \dots & a_{jn} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \tag{7.2}$$

Which, the column with $\vdots$ is the j^{th} column, and the row with $\dots$ is the i^{th} row (and the 0 in between could be different sizes of zero matrices, depending on the amount of entries given). As a result, since $\mathrm{ad}\, A(E_{ij})$ contains no entries of $E_{i'j'}$ (where $i' - j' \ge k$, since everything on these diagonals are all 0), then $\mathrm{ad}\, A(E_{ij}) \in F^{k-1}$, and this proves the claim, showing $\mathrm{ad}\, A \cdot F^k \subset F^{k-1}$.

As a result, since $F^{-n} = 0$ (since no entries satisfy this relation), so $(\mathrm{ad}\, A)^{2n} = 0$ (since for $F^n = \mathrm{End}(V)$, one has $(\mathrm{ad}\, A)^{2n} \cdot F^n \subset F^{n-2n} = F^{-n} = 0$ by induction). So, $\mathrm{ad}\, A$ is nilpotent operator on $\mathrm{End}(V)$.